

# Electromagnetism Learning Plan

Learning-Note

2 October 2026

## Learning method

Build physical meaning before manipulating formulas. For each concept, identify what can be observed or measured, state the mathematical definition, and distinguish it from a physical law or theorem. Stay within the requested concept and advance only after the learner confirms understanding. A planned subject is not a completed lesson.

## Current state

- Flux, divergence, and curl: the learner has confirmed understanding of their definitions and their general purpose. The first note records physical interpretations, applications, and related laws.
- Electric and magnetic fields: the supplied side-chat summary has been checked and incorporated with assumptions about probes, magnetic dipoles, spin, and energy.
- Impedance and voltage/current reflection formulas: previously discussed and reported mathematically understood. Their physical interpretation still needs development.
- Next requested subject: Maxwell's equations. The learner's motivating question is why an ideal short reverses reflected voltage while incident and reflected currents add.

## Proposed sequence

1. **Maxwell's equations, one law at a time.** Start with the physical observation each law describes. Relate its integral and differential forms using the already understood operations. Explain total versus free charge/current and when  $\mathbf{D}$  and  $\mathbf{H}$  are useful.
2. **Displacement current and charge conservation.** Examine a charging capacitor, compare surfaces sharing a boundary, and explain why the time-dependent electric-field term is needed.
3. **Fields at conductors and energy flow.** Introduce surface charge, surface current, ideal-conductor boundary conditions, the Poynting vector, and field energy with explicit assumptions.
4. **Transmission lines as guided fields.** Connect a TEM line's transverse electric field to voltage, magnetic circulation to current, and field storage to distributed capacitance and inductance.
5. **Reflection at an ideal short and open.** Fix current and voltage reference directions. Explain how the conductor responds, how fields superpose, and how backward energy propagation can coexist with a reflected current of the same sign. Distinguish incident, reflected, and total quantities.
6. **Apply the picture to the AN720 antenna.** First identify the exact model, feed, conductor connections, substrate, and relevant geometry using verified documentation and photographs. Then connect current/charge distributions, resonance, matching, polarization, and radiation to the confirmed structure.

## Repository organization and updates

Each subject has its own folder. The current folder is `Electromagnetism-Foundations/`, containing `main.tex` and `main.pdf`. Repository documents are `Documents/plan.tex`, `Documents/plan.pdf`, `Documents/changes.tex`, and `Documents/changes.pdf`.

For the next lesson, extend the foundations note when material directly develops it; create a separate topic folder when the subject merits an independent note. Append a dated change entry and regenerate affected PDFs. Use stable filenames without release numbers or duplicate revision folders.

Only `.tex` and `.pdf` files are intended for Git. Auxiliary outputs and unrelated files are excluded using a repository-local Git exclude rule. Git configuration is not part of an Overleaf upload. Each document uses ordinary pdfLaTeX packages and compiles independently; no external bibliography is needed.